Lab 8: Exception Handling & File Processing

Exercise 1 (1 pts)
Student Grade File Manager
Learning Focus

- File Reading/Writing
- Dictionary
- Exception Handling

Problem Statement
Cho file: students.txt
Có nội dung:
SE001,John,85
SE002,Mary,90
SE003,David,70
SE004,Anna,88

Requirements

1. read file
2. Save to dictionary

Calculate:

- Average grade
- Highest grade
- Lowest grade

Output to file: report.txt

Exercise 2 (1 pts)
Duplicate Student Detection
Learning Focus

- Set
- File Processing
- Data Cleaning

Input File: student_ids.txt
SE001
SE002
SE003
SE001
SE005

SE002
SE006

Requirements
Read file to identify:
Unique IDs, Duplicate IDs
Save result: duplicates.txt

Exercise 3 (1 pts)
Inventory Recovery System
Learning Focus

- Nested Dictionary
- Exception Handling
- Data Persistence

Input File: inventory.csv
P001,Laptop,5,1000
P002,Mouse,20,10
P003,Keyboard,10,25

Requirements
Load file into dictionary.
Calculate:
Total Inventory Value: $quantity \times price$
Generate: inventory_report.txt

Exercise 4 (1 pts)
Course Enrollment Analyzer
Learning Focus

- Set Operations
- File Processing
- Data Analysis

Input Files:
python.txt
SE001
SE002
SE003
SE004

data_science.txt
SE003
SE004
SE005
SE006

Requirements

- Read both files.
- Store using sets.
- Find:
- Students in both courses.
- Students only in Python
- Students only in Data Science
- Total unique students

Save Report: course_analysis.txt

Exercise 5 (1 pts)
Student Ranking Persistence System
Learning Focus

- ✓ List
- ✓ Dictionary
- ✓ Sorting
- ✓ File Saving

Input File: grades.csv
SE001,John,85
SE002,Mary,92
SE003,David,76
SE004,Anna,88
SE005,Tom,65

Requirements:
Sort descending by grade.
Generate ranking.
Classification:

Grade	Classification
>=90	Excellent
>=80	Good
>=65	Average
else	Weak

Save Result: ranking_report.txt

Expected Output:

Rank 1 - Mary - 92 - Excellent
Rank 2 - Anna - 88 - Good
Rank 3 - John - 85 - Good
Rank 4 - David - 76 - Average
Rank 5 - Tom - 65 – Average

Exercise 6 – Large Dataset File Processing
Points: 5 pts
Main Focus: File processing, exception handling, list, dictionary, set
Suggested Dataset
Use a large CSV dataset from Kaggle, for example: **New York City Taxi Trip Duration Dataset**
Use file: train.csv (link here: https://www.kaggle.com/competitions/nyc-taxi-trip-duration/data)
(Dataset này liên quan đến dữ liệu chuyến đi taxi ở New York, gồm thời gian đón khách, tọa độ, số lượng hành khách và thời lượng chuyến đi. Kaggle mô tả bài toán là dự đoán tổng thời gian chuyến đi taxi tại New York)

Problem Statement
Write a Python program to process a large CSV file without using Pandas.
Students must read the file line by line and store selected information using Python data structures learned in Lab 7.

Requirements
Students must:

1. Read the CSV file using open()
2. Process file line by line, not load the whole file at once
3. Store useful information using:
   - ✓ dictionary
   - ✓ set
   - ✓ list
4. Handle exceptions
5. Generate a summary report file

Required Analysis
Students need to calculate:
1. Total number of trips
2. Number of unique vendors
3. Average trip duration
4. Maximum trip duration

5. Trips by passenger count

Required Output File

Generate: large_dataset_report.txt

Expected format:

```
===== LARGE DATASET REPORT =====
Total Trips: 1458644
Unique Vendors: 2
Average Trip Duration: 959.49 seconds
Longest Trip ID: id123456
Longest Trip Duration: 3526282 seconds

Trips by Passenger Count:
1 passenger(s): 1033540 trips
2 passenger(s): 210318 trips
3 passenger(s): 59896 trips
…
```